SUMMARY

(1) Definite integral as the limit of a sum

(i) $\int_a^b f(x)dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{r=1}^{n} f\left(a+(b-a)\frac{r}{n}\right)$

(ii) $\int_0^1 f(x)dx = \lim_{n\to\infty} \frac{1}{n} \sum_{r=0}^{n} f\left(\frac{r}{n}\right) = \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} f\left(\frac{r}{n}\right).$

(2) Properties of definite integrals

(i) $\int_a^b f(x)\,dx = \int_a^b f(u)\,du$

(ii) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(iii) $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

(iv) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(v) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

(vi) $\int_0^{2a} f(x)\,dx = \int_0^a [f(x)+f(2a-x)]\,dx.$

(vii) If $f(x)$ is an even function, then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx.$

(ix) If $f(x)$ is an odd function, then $\int_{-a}^{a} f(x)\,dx = 0.$

(x) If $f(2a-x) = f(x)$, then $\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx.$

(xi) If $f(2a-x) = -f(x)$, then $\int_0^{2a} f(x)\,dx = 0.$

(xii) $\int_0^a x\,f(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx$ if $f(a-x) = f(x)$

(3) Bernoulli's Formula

$$\int uvdx = uv_{(1)} - u^{(1)}v_{(2)} + u^{(2)}v_{(3)} - u^{(3)}v_{(4)} + \cdots.$$

(4) Reduction Formulae

(i) $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \int_0^{\frac{\pi}{2}} \cos^n x\,dx = \begin{cases} \frac{(n-1)}{n}\times\frac{(n-3)}{(n-2)}\times\cdots\times\frac{1}{2}\times\frac{\pi}{2}, & \text{if } n = 2,4,6,\cdots \\ \frac{(n-1)}{n}\times\frac{(n-3)}{(n-2)}\times\cdots\times\frac{2}{3}, & \text{if } n = 3,5,7,\cdots \end{cases}$

(ii) If n is even and m is even,

$$\int_0^{\frac{\pi}{2}} \sin^m x\,\cos^n x\,dx = \frac{(n-1)}{(m+n)}\frac{(n-3)}{(m+n-2)}\frac{(n-5)}{(m+n-4)}\cdots\frac{1}{(m+2)}\frac{(m-1)}{m}\frac{(m-3)}{(m-2)}\frac{(m-5)}{(m-4)}\cdots\frac{1}{2}\frac{\pi}{2}$$

(iii) If n is odd and m is any positive integer (even or odd), then

$$\int_0^{\frac{\pi}{2}} \sin^m x\,\cos^n x\,dx = \frac{(n-1)}{(m+n)}\frac{(n-3)}{(m+n-2)}\frac{(n-5)}{(m+n-4)}\cdots\frac{2}{(m+3)}\frac{1}{(m+1)}.$$

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